

PHYSICS "SURE-SHOT" TREND TEST

Curated from 8 High-Frequency Sets (2023-2025 Analysis)

Time: 3 Hours | Max Marks: 70 | Syllabus: 2025-26 Certified

SECTION A: TOP REPEATED MCQs (1 Mark)

[Repeated in 3 Papers: 2025 S1, 2025 S2, 2023]

Q1. Two nuclei have mass numbers in ratio 27:64. The ratio of their nuclear densities is: (1)

- (A) 27:64
- (B) 1:1
- (C) 3:4
- (D) 9:16

[Repeated in 3 Papers: 2025 S1, 2025 S2, 2023]

Q2. The radius of the n -th orbit in Bohr's model of H-atom is proportional to: (1)

- (A) n
- (B) n^2
- (C) $1/n$
- (D) $1/n^2$

[Repeated in 3 Papers: 2025, 2024]

Q3. In a forward biased p-n junction diode: (1)

- (A) Barrier height increases
- (B) Depletion width decreases
- (C) Depletion width increases
- (D) No change occurs

[Repeated in 3 Papers: 2024, 2025]

Q4. Slope of Stopping Potential (V_0) vs Frequency (ν) graph represents: (1)

- (A) h
- (B) h/e
- (C) e/h
- (D) Work function

[Repeated in 2 Papers: 2025 S1, 2024]

Q5. EM waves used in water purifiers to kill germs are: (1)

- (A) Infrared
- (B) UV rays
- (C) X-rays
- (D) Microwaves

[Repeated in 2 Papers: Assertion-Reason Trend]

Q6. Assertion: EM waves don't require a medium.

Reason: EM waves are transverse in nature.

(1)

SECTION B: VERY SHORT ANSWER (2 Marks)

[Repeated in 2 Papers: 2025 S1, S2]

Q7. Draw energy band diagrams of n-type and p-type semiconductors at $T > 0\text{ K}$. (2)

[Repeated in 3 Papers: Current & Force]

Q8. Two long parallel wires carry current in same direction. State nature of force and formula for force per unit length. (2)

[Repeated in 2 Papers: 2024, 2025]

Q9. Calculate Binding Energy per nucleon for a Helium nucleus (Mass provided in exam data). (2)

SECTION C: SHORT ANSWER & DERIVATIONS (3 Marks)

[RANKED #1 REPEATED DERIVATION]

Q10. Using Bohr's postulates, derive expression for radius of n-th orbit. Show $r \propto n^2$. (3)

[Huygens Trend]

Q11. Define wavefront. Use Huygens' Principle to verify Snell's Law of Refraction. (3)

[Galvanometer Trend]

Q12. How is a galvanometer converted into a voltmeter? Explain current sensitivity. (3)

[Dual Nature Trend]

Q13. Plot graph of Stopping Potential vs Frequency. Determine Work Function and Planck's constant from it. (3)

SECTION D: CASE STUDIES (4 Marks)

[Electricity Trend]

Q14. Case Study: Drift Velocity. (i) Establish relation between Current and vd . (ii) Effect of Temp on resistivity for conductors vs semiconductors. (4)

[Optics Trend]

Q15. Case Study: Lens Maker. Use Lens Maker formula to find focal length. Explain change in focal length when lens is immersed in water. (4)

SECTION E: LONG ANSWER (5 Marks)

[REPEATED IN 4/4 YEARS - DO NOT SKIP]

Q16. Ray Optics: Draw labelled ray diagram of an Astronomical Telescope for normal adjustment. Derive expression for Magnifying Power. (5)

[Semiconductor Trend]

Q17. Full Wave Rectifier: Explain working with circuit diagram. Draw input and output waveforms. (5)

[Electrostatics Trend]

Q18. Capacitance: Derive expression for capacitance of a parallel plate capacitor with a dielectric slab of thickness $t < d$. (5)

GOLDEN REPEAT SUMMARY (SURE-SHOT):

1. **Ray Optics (Telescope):** 4/4 Years. (5 Marks)
2. **Semiconductors (Rectifier):** 3/4 Years. (5 Marks)
3. **Bohr's Model (Radius):** 3/4 Years. (3 Marks)
4. **Nuclear Density (1:1):** 3/4 Years. (1 Mark)
5. **Drift Velocity:** 3/4 Years. (4 Marks Case Study)